Ayush Mishra

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EDUCATION

International Institute of Information Technology, Bangalore **August 2023 – July 2028**
Integrated Master of Technology in Computer Science and Engineering *CGPA 3.55/4.0*

EXPERIENCE

COMET Lab **May 2026 – Present**
Summer Intern *IIIT Bangalore*

- Engineered a reproducible **QoS–QoE video streaming dataset** using **Mininet**, **SRT**, and **VMAF** by integrating per-second bandwidth, delay, jitter, and packet-loss measurements with QoE labels to generate **500+ samples**.
- Designed a **4-zone Mininet topology** across 6 network profiles; migrated from raw UDP to **SRT** to eliminate data-moshing artifacts and computed per-second VMAF metrics aligned with QoS data.

PROJECTS

Concurrent Course Registration System / ***C, POSIX Sockets, pthreads*** **April 2026 – May 2026**

- Built a multithreaded TCP client-server course registration system in **C** using POSIX sockets and pthreads, supporting 3 role-based workflows (admin, faculty, student) with binary file-backed persistence across **250+ users**.
- Stress-tested under **250+ concurrent clients on localhost**, processing **12,000+ requests** at **760+ ops/sec (view/enroll/drop/logout)** with **0% error rate**; server memory footprint held at under **10 MB**.
- Prevented a **TOCTOU race condition** causing data corruption by implementing a two-layer locking strategy (pthread mutex + fcntl write lock); validated using a **100-thread barrier test** across **160,000+** lock acquisitions with **0 high-contention events** (> 1 ms lock wait).

Linux Performance Tuning / ***C, Linux, Performance Profiling Tools*** **January 2026 – May 2026**

- Optimized Linux CPU performance under **6-process contention** using CPU affinity and real-time scheduling (`SCHED_RR`), reducing execution time by **80.7%** (**43.7s to 8.45s**).
- Isolated two independent effects: CPU affinity reduced execution time by **49%** and `SCHED_RR` reduced context switches by **99%** (**2449 to 24**); validated across **30+ benchmark runs** via Welch's t-test ($\boldsymbol{p < 0.001}$).

Visual Product Search Engine / ***Python, FAISS, HNSW*** **April 2026 – May 2026**

- Developed a multimodal vector retrieval system using **FAISS** and **HNSW**-based approximate nearest neighbor indexing over **55k+** image-text embeddings, with category-aware filtering and exact cosine re-ranking across **15k+** product images.
- Evaluated retrieval quality using Recall@K, mAP@K, and NDCG@K ranking metrics, achieving a **Recall@15 of 0.814** on **1,330+ query images**; built an interactive **Streamlit** demo with crop confirmation and real-time Top-K visual search.

Multi-Pipeline ETL Framework / ***MongoDB, PostgreSQL, Distributed ETL*** **April 2026 – May 2026**

- Implemented the **MongoDB aggregation pipeline** in a 4-backend ETL framework processing **3.4M+ NASA HTTP log records** across **35 batches** with batch-streaming, regex-based CLF parsing, and malformed record isolation.
- Designed **3 analytical queries** (daily traffic, top resources, hourly error analysis) with results and execution metadata stored in **PostgreSQL**; MongoDB achieved the lowest overall runtime at **141,896 ms** across all 4 pipeline backends.

TECHNICAL SKILLS

Languages: C++, C, Python, Java, SQL, JavaScript
Web Development: Node.js, Express.js, React, MongoDB
Systems: Operating Systems, Linux, System Programming, PostgreSQL, POSIX Sockets, Multithreading, Concurrency
Developer Tools: Git, GitHub, Postman, VS Code, IntelliJ IDEA

RELEVANT COURSEWORK

Data Structures, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering, Machine Learning

MENTORSHIP

Teaching Assistant – Computer Architecture **January 2026 – May 2026**
*Mentored **150+ students** and conducted **16+ tutorial sessions*** *IIIT Bangalore*